Resolved Axisymmetric Tensor Analysis of Dragonfly 44

Stellar Second and Fourth-Moment Dynamics in CPTG

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Abstract

Dragonfly 44 is an ultra-diffuse galaxy whose low stellar density and substantial pressure support make it a demanding mass-discrepancy system. This paper analyzes its resolved stellar kinematics in Curvature Polarization Transport Gravity (CPTG), a geometric framework in which ordinary baryons remain the material source while curvature polarization and transport provide the additional gravitational response. The measured Sérsic structure is deprojected as an oblate stellar source, the baryonic potential is evaluated in that geometry, the CPTG scalar galaxy response is imposed on the equatorial plane, and its off-plane extension is constructed on baryonic equipotential surfaces so that a scalar effective potential exists. One object-level structural acceleration is determined from the published $33 \pm 3\,\mathrm{km\,s^{-1}}$ circularized half-light measurement; no radial bin receives an independent value. The resulting field is propagated through the stationary axisymmetric Jeans equations and the projected observation operator for the nine KCWI apertures. For the central $33\,\mathrm{km\,s^{-1}}$ half-light normalization, the edge-on neutral branch gives $a_\star = 5.294 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, places every resolved second-moment datum within 1.47σ, and gives the descriptive asymmetric statistic $\chi^2_{\mathrm{diag}} = 9.21$. Repeating the whole-galaxy normalization at 30 and $36\,\mathrm{km\,s^{-1}}$ gives maximum pointwise pulls of 1.16σ and 2.45σ, respectively, without radial renormalization. The same calculation with the CPTG response removed gives $\chi^2_{\star,\mathrm{diag}} = 539.66$ and only $6.14\,\mathrm{km\,s^{-1}}$ inside the circularized half-light aperture. A minimal fourth-order Jeans closure gives slight negative line-of-sight excess kurtosis, whereas nonnegative three-integral Schwarzschild-style orbit superpositions in the fixed CPTG potential reproduce the measured $h_4 = 0.13 \pm 0.05$ while preserving the tracer-density and second-moment constraints. The solved continuous field gives Structural Mode $N = 1.655926168$ on the exact parent interval from $R/5$ to the outermost modeled KCWI semi-major coordinate.

1 Introduction

Dragonfly 44 is a large, low-surface-brightness ultra-diffuse galaxy in the Coma cluster and is widely regarded as an archetypal "failed galaxy": under the conventional halo-based interpretation, it assembled a disproportionately massive halo and rich globular-cluster population while forming far less field-star mass than expected for that halo [1, 2]. Its substantial pressure support and gravitational mass discrepancy therefore make it an unusually direct test of whether the observed stellar source can account for the resolved dynamics without an added non-luminous matter component.

The kinematic constraints adopted here are the spatially resolved Keck Cosmic Web Imager (KCWI) measurements of van Dokkum et al. [3]. They give the luminosity-weighted half-light second moment

$$\sigma_{1/2} = 33 \pm 3 \ \mathrm{km\,s^{-1}}, \tag{1}$$

with $V_{\max}/\langle\sigma\rangle < 0.12$ at 90% confidence and a spatially averaged Gauss–Hermite coefficient $h_4 = 0.13 \pm 0.05$. The resolved second-moment profile rises from approximately $26\,\mathrm{km\,s^{-1}}$ in the inner galaxy to approximately $41\,\mathrm{km\,s^{-1}}$ at the outermost aperture. Because ordered rotation is negligible, the relevant observable is a projected stellar stress field rather than a circular-speed curve.

Curvature Polarization Transport Gravity (CPTG) is a geometric gravitational framework in which ordinary baryonic matter remains the material source and the additional galaxy-scale gravitational response is carried by curvature polarization and curvature transport rather than by an added dark-matter stress-energy component [11]. In schematic covariant form,

$$G_{\mu\nu} + \Pi^{\mathrm{pol}}_{\mu\nu} + \Theta^{\mathrm{trans}}_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{2}$$

Here $T_{\mu\nu}$ is the stress-energy of ordinary matter, $\Pi^{\mathrm{pol}}_{\mu\nu}$ is the curvature-polarization response generated by that source, and $\Theta^{\mathrm{trans}}_{\mu\nu}$ describes the directional organization and transport of the geometric response. The latter two terms belong to the gravitational sector and are not additional matter components.

For Dragonfly 44 the calculation therefore proceeds from the measured stellar source to the gravitational field and then to the observed line-of-sight moments. The Sérsic light profile is deprojected as an oblate stellar density; its baryonic potential is computed directly; the CPTG equatorial response is extended off the plane through baryonic equipotential surfaces; one galaxy-level structural acceleration is fixed by the integrated half-light support condition; and the stationary axisymmetric Jeans equations propagate that field into the projected second moments of all nine KCWI apertures. The same fixed potential is then carried into the fourth-moment and three-integral LOSVD calculation. Structural Mode N is evaluated only after the dynamical field is solved; it is a dimensionless post-solution coordinate that identifies where the radial variation of the solved CPTG field is concentrated relative to the resolved outer scale.

2 Observed Structure and Radius Convention

All physical radii are evaluated on the $D = 100\,\mathrm{Mpc}$ distance scale adopted for the KCWI analysis [3].

The adopted stellar structure is

$$R_{e,\mathrm{maj}} = 4.7\ \mathrm{kpc}, \qquad n = 0.94, \qquad q_{\mathrm{proj}} = 0.68, \tag{3}$$

with a baseline stellar mass [1, 3]

$$M_\star = 3.0 \times 10^8\, M_\odot. \tag{4}$$

The source-sensitivity calculations span 3.0×10^8–$4.5 \times 10^8\, M_\odot$; the upper endpoint is an analysis bound rather than an additional measured stellar mass. The KCWI extraction ellipses use approximately $q_{\mathrm{ap}} = 0.69$ [3].

The source table reports luminosity-weighted aperture coordinate R and states that the corresponding major-axis coordinate is approximately $1.20R$. The present calculation adopts the numerical convention

$$R_{\mathrm{maj}} = \frac{R}{0.83}. \tag{5}$$

Both coordinates are retained explicitly. The source relation is approximate; a direct $R_{\mathrm{maj}} = 1.20R$ sensitivity calculation is reported below. The published observable is

$$\sigma_{\mathrm{eff}}(R) = \sqrt{\sigma^2(R) + v^2(R)}. \tag{6}$$

The photometric circularized half-light radius is

$$R_{e,\mathrm{circ}} = R_{e,\mathrm{maj}}\sqrt{q_{\mathrm{proj}}} = 3.8757193\ \mathrm{kpc}. \tag{7}$$

The $33 \pm 3\,\mathrm{km\,s^{-1}}$ normalization is applied to the projected circular aperture $r_{\mathrm{sky}} \le R_{e,\mathrm{circ}}$.

3 Stellar Source and Baryonic Field

The projected stellar mass surface density is represented by

$$\Sigma(s) = \Sigma_e \exp\left[-b_n\left(\left(\frac{s}{R_{e,\mathrm{maj}}}\right)^{1/n} - 1\right)\right]. \tag{8}$$

For intrinsic oblate axis ratio q_{int}, the density is obtained by Abel deprojection,

$$\nu(m) = -\frac{q_{\mathrm{proj}}}{\pi q_{\mathrm{int}}}\int_m^\infty \frac{d\Sigma/ds}{\sqrt{s^2 - m^2}}\, ds, \qquad m^2 = R^2 + \frac{z^2}{q_{\mathrm{int}}^2}. \tag{9}$$

For a density stratified on similar oblate spheroids, define

$$m^2(u) = \frac{R^2}{1+u} + \frac{z^2}{q_{\rm int}^2 + u}, \qquad H(m) = 2\int_m^\infty \nu(t)t\,dt. \tag{10}$$

The baryonic potential is

$$\Phi_\star(R,z) = -\pi G q_{\rm int} \int_0^\infty \frac{H[m(u)]}{(1+u)\sqrt{q_{\rm int}^2+u}}\,du, \tag{11}$$

with positive potential-gradient components

$$\frac{\partial \Phi_\star}{\partial R} = 2\pi G q_{\rm int} R \int_0^\infty \frac{\nu[m(u)]}{(1+u)^2\sqrt{q_{\rm int}^2+u}}\,du, \tag{12}$$

$$\frac{\partial \Phi_\star}{\partial z} = 2\pi G q_{\rm int} z \int_0^\infty \frac{\nu[m(u)]}{(1+u)(q_{\rm int}^2+u)^{3/2}}\,du. \tag{13}$$

The stellar density is normalized directly to $M_\star$; no stellar-source weight is introduced.

4 Conservative Axisymmetric CPTG Reduction

The CPTG galaxy magnitude equation is

$$g = g_{\rm bar} + \frac{\sqrt{a_\star g_{\rm bar}} + g_{\rm geom}(r)}{\left[1+(g/a_\star)^{123/50}\right]^{11/50}}, \tag{14}$$

where

$$g_{\rm geom}(r) = \frac{1}{2} a_\star \frac{(r/R_c)^{7/5}}{1+(r/R_c)^{12/5}}, \qquad R_c = \frac{48\pi c^2}{a_\star}. \tag{15}$$

For positive $a_\star$ and $g_{\rm bar}$ the implicit root is unique because the corresponding root function has strictly positive derivative. Numerical root residuals and monotonic derivatives are retained in the evidence package.

To extend the equatorial magnitude law into the oblate stellar body while preserving a scalar potential, define the baryonic potential label

$$u = \Phi_\star(R,z) - \Phi_\star(0,0). \tag{16}$$

For each u, let $R_{\rm eq}(u)$ denote the equatorial radius with the same baryonic potential label. Solve Eq. (14) there and define

$$\mathcal{A}(u) = \frac{g_{\rm CPTG,eq}[R_{\rm eq}(u)]}{g_{\rm bar,eq}[R_{\rm eq}(u)]}, \qquad u > 0. \tag{17}$$

The axisymmetric CPTG field is then supplied through

$$\nabla\Phi_{\rm CPTG} = \mathcal{A}(u)\nabla\Phi_\star. \tag{18}$$

Since $\nabla u = \nabla \Phi_\star$,

$$\nabla \times [\mathcal{A}(u)\nabla\Phi_\star] = \mathcal{A}'(u)\nabla u \times \nabla\Phi_\star = 0. \tag{19}$$

Thus a scalar effective potential exists,

$$\Phi_{\rm CPTG}(u) = \Phi_{\rm CPTG}(0) + \lim_{\epsilon\to 0^+} \int_\epsilon^u \mathcal{A}(s)\, ds. \tag{20}$$

At the exact center both equatorial accelerations vanish, so the ratio in Eq. (17) is not evaluated at $u = 0$. For the present $n < 1$ Sérsic source the central baryonic field is regular, $g_{\rm bar} = O(r)$, while the low-acceleration CPTG response gives $g_{\rm CPTG} = O(r^{1/2})$. Hence $\mathcal{A} = O(r^{-1/2})$, but the physical force $\mathcal{A}\nabla\Phi_\star = O(r^{1/2}) \to 0$. Since $u = O(r^2)$, $\mathcal{A}(u) = O(u^{-1/4})$ and the lower endpoint in Eq. (20) is integrable.

This axisymmetric reduction is used throughout the calculation. It preserves the CPTG scalar galaxy law exactly on the equatorial plane and introduces no additional fitted coefficient. It is not asserted to be the complete arbitrary-geometry form of the parent directional transport tensor.

For the edge-on neutral solution below,

$$R_c = 829.7314 \text{ Gpc}, \tag{21}$$

so $g_{\rm geom}$ is negligible throughout the measured stellar body; for example it is approximately $7.5 \times 10^{-22}\,\mathrm{m\,s^{-2}}$ at $4.7\,\mathrm{kpc}$.

5 Axisymmetric Stellar Second Moments

The stationary axisymmetric Jeans equations are [4, 5]

$$\frac{\partial(\nu\overline{v_R^2})}{\partial R} + \frac{\partial(\nu\overline{v_R v_z})}{\partial z} + \nu\left[\frac{\overline{v_R^2} - \overline{v_\phi^2}}{R} + \frac{\partial\Phi_{\rm CPTG}}{\partial R}\right] = 0, \tag{22}$$

$$\frac{\partial(\nu\overline{v_R v_z})}{\partial R} + \frac{\partial(\nu\overline{v_z^2})}{\partial z} + \frac{\nu\overline{v_R v_z}}{R} + \nu\frac{\partial\Phi_{\rm CPTG}}{\partial z} = 0. \tag{23}$$

The edge-on neutral solution adopts cylindrical alignment,

$$\overline{v_R v_z} = 0, \tag{24}$$

and the neutral meridional state

$$\beta_z \equiv 1 - \frac{\sigma_z^2}{\sigma_R^2} = 0. \tag{25}$$

The observed upper limit on ordered rotation motivates $\overline{v}_\phi = 0$ for the edge-on neutral solution. The vertical pressure is therefore

$$\nu\sigma_z^2(R,z) = \int_z^\infty \nu(R,z')\frac{\partial\Phi_{\rm CPTG}}{\partial z'}\, dz', \tag{26}$$

and the azimuthal second moment follows from the radial equation,

$$\overline{v_\phi^2} = \sigma_R^2 + R\frac{\partial \Phi_{\mathrm{CPTG}}}{\partial R} + \frac{R}{\nu}\frac{\partial(\nu\sigma_R^2)}{\partial R}. \tag{27}$$

Because $\overline{v}_\phi = 0$, the azimuthal second moment is the azimuthal velocity dispersion squared in this branch. Any geometry/stress branch is excluded if a non-negligible active tracer region develops a negative azimuthal second moment.

6 Projection and Structural Normalization

At each sky position the intrinsic tensor is projected as

$$\overline{v_{\mathrm{los}}^2} = n_i n_j \overline{v_i v_j}. \tag{28}$$

The numerator and surface-density denominator are integrated along the line of sight, convolved with an adopted Gaussian spatial-resolution approximation of FWHM 1.2″, and then integrated over the $q_{\mathrm{ap}} = 0.69$ elliptical KCWI annuli. The model observable is

$$\sigma_{\mathrm{eff},k}^{\mathrm{CPTG}} = \sqrt{\langle v_{\mathrm{los}}^2 \rangle_k}. \tag{29}$$

For each geometry/stress branch, one galaxy-level structural acceleration is determined by

$$\sigma_{\mathrm{CPTG}}(< R_{e,\mathrm{circ}}; a_\star) = 33\ \mathrm{km\,s^{-1}}. \tag{30}$$

For the high-resolution edge-on neutral branch at the central $33\,\mathrm{km\,s^{-1}}$ half-light normalization,

$$a_\star = 5.2935187 \times 10^{-10}\ \mathrm{m\,s^{-2}}. \tag{31}$$

The nine radial measurements are not independently normalized. The resolved comparison therefore measures shape, projection, and field organization after the single integrated support condition has been imposed. The $33\,\mathrm{km\,s^{-1}}$ solution is the central normalization result; the quoted $\pm 3\,\mathrm{km\,s^{-1}}$ measurement range is propagated separately as a sensitivity calculation rather than added as an independent likelihood term.

7 Resolved Dragonfly 44 Result

The reported asymmetric errors in v and σ are propagated to σ_{eff} using first-order error propagation. For a compact descriptive diagnostic, define

$$\chi^2_{\mathrm{diag}} = \sum_i \frac{(\sigma_{\mathrm{CPTG},i} - \sigma_{\mathrm{obs},i})^2}{s_i^2}, \tag{32}$$

where s_i is the propagated upper or lower uncertainty in the direction of the residual. Because a complete inter-bin covariance matrix is not available, Eq. (32) is not treated as a complete likelihood.

R	R_{maj}	Observed σ_{eff}	CPTG	Stellar-only	Residual	Pull
(kpc)	(kpc)	(km s^{-1})	(km s^{-1})	(km s^{-1})	(km s^{-1})	
0.23	0.277	$26.101^{+4.400}_{-3.500}$	30.847	5.832	+4.746	+1.079
0.49	0.590	$26.916^{+4.083}_{-3.389}$	31.234	5.913	+4.318	+1.058
0.79	0.952	$26.681^{+4.380}_{-2.892}$	31.787	6.025	+5.106	+1.166
1.13	1.361	$31.808^{+3.299}_{-2.900}$	32.348	6.129	+0.541	+0.164
1.53	1.843	$29.104^{+3.400}_{-2.400}$	32.889	6.211	+3.785	+1.113
1.94	2.337	$29.508^{+3.000}_{-2.600}$	33.327	6.258	+3.818	+1.273
2.55	3.072	$29.303^{+3.100}_{-2.500}$	33.832	6.273	+4.529	+1.461
3.62	4.361	$34.401^{+3.800}_{-3.600}$	34.456	6.205	+0.055	+0.014
5.13	6.181	$40.631^{+7.836}_{-7.543}$	35.062	6.008	-5.569	-0.738

Table 1: Resolved Dragonfly 44 comparison. R is the published aperture coordinate and $R_{maj} = R/0.83$ is the semi-major coordinate used by the model annuli. Pulls use the asymmetric uncertainty in the direction of the CPTG residual.

The edge-on neutral result is

$$\chi^2_{diag} = 9.20835, \qquad \max_i |\mathrm{pull}_i| = 1.46112. \tag{33}$$

Thus, for the central $33\,\mathrm{km\,s^{-1}}$ half-light normalization, every resolved KCWI aperture lies within 1.47σ of the edge-on neutral CPTG tensor prediction.

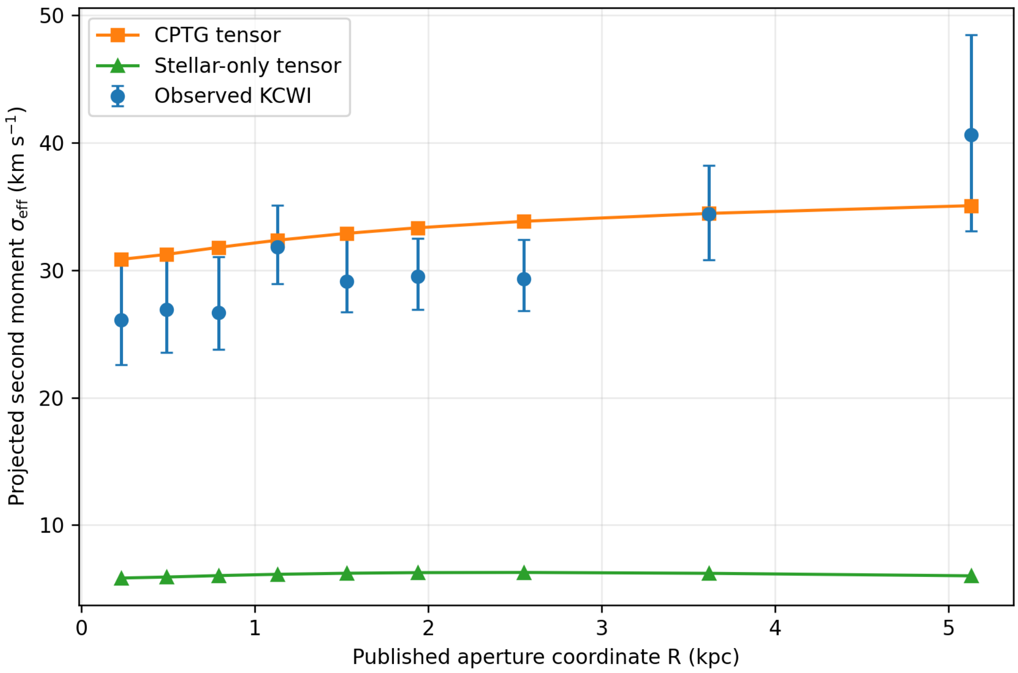

Figure 1: Observed KCWI projected second moments compared with the conservative CPTG tensor solution and the identical stellar-only tensor control. The model annuli use Eq. (5); the horizontal coordinate shown is the published R.

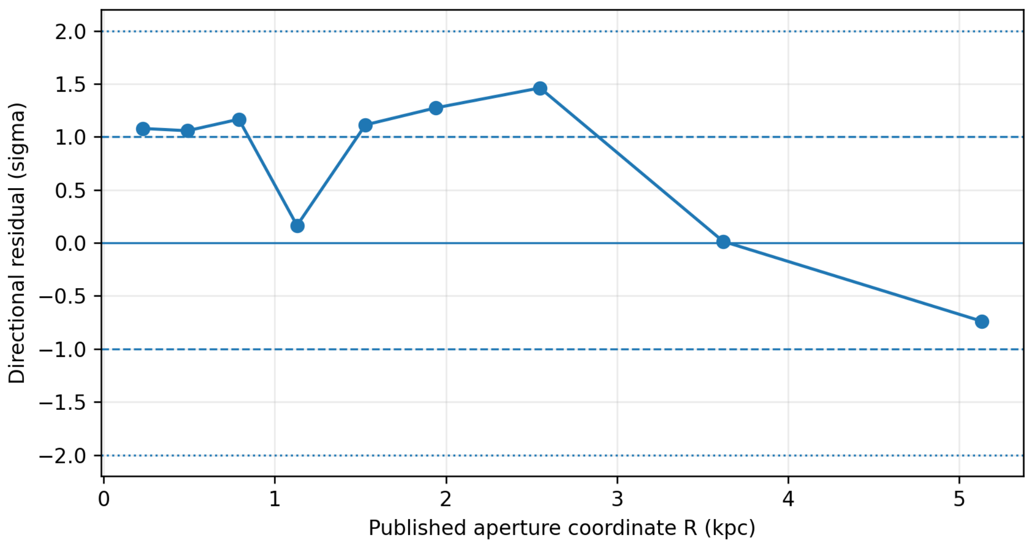

Figure 2: Directional residuals of the edge-on neutral CPTG tensor solution in units of the corresponding asymmetric observational uncertainty.

8 Stellar-Only Control

The control calculation retains the same oblate source, Jeans equations, projection, spatial-resolution convolution, circular half-light aperture, and elliptical radial apertures, but supplies only the stellar baryonic field. It gives

$$\sigma_\star(< R_{e,\mathrm{circ}}) = 6.13561\ \mathrm{km\,s^{-1}}, \tag{34}$$

and

$$\chi^2_{\star,\mathrm{diag}} = 539.657. \tag{35}$$

The largest stellar-only pointwise discrepancy is 9.54σ. Axisymmetric tensor geometry by itself therefore does not account for the observed dynamical support under the adopted stellar source.

9 Sensitivity and Numerical Validation

9.1 Half-light normalization

The half-light measurement is $33 \pm 3\,\mathrm{km\,s^{-1}}$. To propagate that normalization uncertainty without treating it as an independent radial datum, the complete edge-on neutral calculation is repeated at 30, 33, and $36\,\mathrm{km\,s^{-1}}$. In each row a single galaxy-level $a_\star$ is re-determined from the corresponding integrated half-light support condition; the nine resolved apertures remain untouched.

Half-light target	$a_\star$ (m s^{-2})	χ^2_{diag}	max |pull|
$30\,\mathrm{km\,s^{-1}}$	3.562×10^{-10}	3.699	1.163
$33\,\mathrm{km\,s^{-1}}$	5.294×10^{-10}	9.208	1.461
$36\,\mathrm{km\,s^{-1}}$	7.583×10^{-10}	26.454	2.454

Table 2: Sensitivity to the quoted half-light normalization uncertainty. These rows are alternative whole-galaxy normalizations, not additional independent likelihood terms.

Across the quoted $\pm 3\,\mathrm{km\,s^{-1}}$ interval, the inferred structural acceleration spans 3.562×10^{-10}–$7.583 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. The largest resolved pointwise pull remains below 2.46σ. The central $33\,\mathrm{km\,s^{-1}}$ row is used for the primary profile and for the fixed potential carried into the higher-moment calculation.

9.2 Inclination

For an oblate system,

$$q_{\mathrm{int}}^2 = \frac{q_{\mathrm{proj}}^2 - \cos^2 i}{\sin^2 i}. \tag{36}$$

The neutral branch was evaluated at 180 radial nodes and 112 force nodes at $i = 50^\circ, 60^\circ, 70^\circ, 80^\circ, 90^\circ$ without selecting an inclination from the nine radial residuals.

i	q_{int}	$a_\star$ (m s^{-2})	χ^2_{diag}	max |pull|
50°	0.290	9.375×10^{-10}	9.306	1.771
60°	0.532	6.242×10^{-10}	9.516	1.518
70°	0.625	5.592×10^{-10}	9.428	1.473
80°	0.668	5.352×10^{-10}	9.368	1.457
90°	0.680	5.286×10^{-10}	9.349	1.453

Table 3: Neutral $\beta_z = 0$ inclination family evaluated at 180 radial nodes and 112 force nodes.

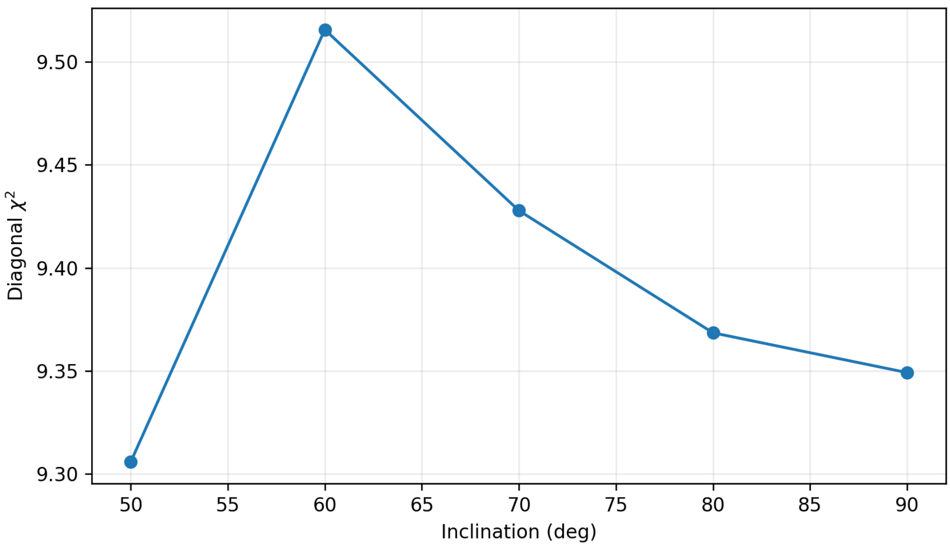

Figure 3: Diagonal residual statistic across the neutral inclination family at 180 radial nodes and 112 force nodes.

The resolved agreement therefore does not require a selected viewing angle. The most face-on allowed point in the stated grid remains below 1.8σ at every resolved aperture.

9.3 Meridional stress

The meridional-stress grid evaluates

$$\beta_z = -0.2,\ 0,\ 0.2,\ 0.4 \tag{37}$$

at each of the five inclinations above. All 20 branches retain positive azimuthal second moments over the active tracer domain. The stronger $\beta_z = 0.4$ cases degrade rather than improving the radial comparison; at the same 180-node/112-force-node resolution the diagonal statistic rises to about 19.6 with a maximum pointwise deviation of about 1.97σ. The tensor closure therefore does not make arbitrary stress structure equally successful.

9.4 Stellar mass

At 180 radial nodes and 112 force nodes, the edge-on neutral branch gives

$M_\star$ ($M_\odot$)	$a_\star$ (m s^{-2})	$\chi^2_{\rm diag}$	max |pull|
3.00×10^8	5.286×10^{-10}	9.349	1.453
3.75×10^8	4.154×10^{-10}	9.367	1.453
4.50×10^8	3.400×10^{-10}	9.386	1.452

Table 4: Stellar-mass sensitivity of the edge-on neutral branch at 180 radial nodes and 112 force nodes.

The structural acceleration responds to the adopted stellar source, while the resolved projected shape changes very little across the stated stellar-population interval.

9.5 Spatial resolution and numerical convergence

The $1.2''$ Gaussian spatial-resolution operator changes the innermost edge-on neutral prediction by $+0.317\,\mathrm{km\,s^{-1}}$ and changes every other aperture by less than $0.10\,\mathrm{km\,s^{-1}}$ in magnitude. This is well below the reported observational uncertainty of each bin. At the 180-node/112-force-node resolution, replacing the adopted $R_{\mathrm{maj}} = R/0.83$ convention by the literal approximate relation $R_{\mathrm{maj}} = 1.20R$ changes χ^2_{diag} from 9.349 to 9.325 and the maximum pull from 1.453 to 1.452, with no change in $a_\star$ at the quoted precision.

The resolution sequence is

Grid	Force nodes	$a_\star$ (m s^{-2})	χ^2_{diag}	max |pull|
120	80	5.2702×10^{-10}	9.626	1.444
180	112	5.2859×10^{-10}	9.349	1.453
240	136	5.2907×10^{-10}	9.267	1.459
320	176	5.2935×10^{-10}	9.208	1.461

Table 5: Numerical convergence of the edge-on neutral calculation.

The structural acceleration changes by about 5.3×10^{-4} fractionally between the last two rows, small compared with the observational scale of the problem.

The Abel source calculation closes its normalization at approximately 3.6×10^{-5} relative level, and the Sérsic reprojection error is also approximately 3.6×10^{-5} over the evaluated radial range. The normalized finite-difference curl diagnostic for Eq. (18) decreases monotonically with source-grid resolution and reaches a 95th-percentile value of 7.23×10^{-4} on the 320-node grid, consistent with the exact continuum identity that the constructed field is curl-free.

10 Fourth-Moment and Three-Integral LOSVD Analysis

The measured Gauss–Hermite coefficient $h_4 = 0.13\pm0.05$ supplies a higher-order constraint beyond the second-moment solution. Van Dokkum et al. obtain this coefficient from the optimally extracted spectrum and assign that extraction a luminosity-weighted radius of $1.3\,\mathrm{kpc}$ [3]. Throughout this section the conservative CPTG potential, tracer density, and structural acceleration determined above remain fixed.

10.1 Minimal fourth-order Jeans diagnostic

Define the even intrinsic fourth moments

$$\begin{aligned} &M_{R4} = \overline{v_R^4}, \qquad M_{z4} = \overline{v_z^4}, \qquad M_{\phi 4} = \overline{v_\phi^4}, \\ &M_{R2z2} = \overline{v_R^2 v_z^2}, \qquad M_{R2\phi 2} = \overline{v_R^2 v_\phi^2}, \qquad M_{\phi 2z2} = \overline{v_\phi^2 v_z^2}. \end{aligned} \tag{38}$$

For this deliberately minimal fourth-order closure, assume the local velocity distribution is even under independent sign reversals of v_R, v_z, and v_ϕ. Odd mixed velocity moments therefore vanish, and the stationary fourth-order Jeans hierarchy contains

$$\frac{\partial(\nu M_{z4})}{\partial z} + 3\nu\sigma_z^2 \frac{\partial \Phi_{\mathrm{CPTG}}}{\partial z} = 0, \tag{39}$$

$$\frac{\partial(\nu M_{R2z2})}{\partial z} + \nu\sigma_R^2 \frac{\partial \Phi_{\mathrm{CPTG}}}{\partial z} = 0, \tag{40}$$

$$\frac{\partial(\nu M_{\phi 2z2})}{\partial z} + \nu\overline{v_\phi^2} \frac{\partial \Phi_{\mathrm{CPTG}}}{\partial z} = 0, \tag{41}$$

with radial relations

$$\frac{1}{R}\frac{\partial(R\nu M_{R4})}{\partial R} - \frac{3\nu}{R} M_{R2\phi 2} + 3\nu\sigma_R^2 \frac{\partial \Phi_{\mathrm{CPTG}}}{\partial R} = 0, \tag{42}$$

$$\frac{1}{R}\frac{\partial(R\nu M_{R2\phi 2})}{\partial R} + \frac{2\nu}{R} M_{R2\phi 2} - \frac{\nu}{R} M_{\phi 4} + \nu\overline{v_\phi^2} \frac{\partial \Phi_{\mathrm{CPTG}}}{\partial R} = 0. \tag{43}$$

The second-order condition $\sigma_R^2 = \sigma_z^2$ does not uniquely close this hierarchy [7]. As a deliberately minimal diagnostic, impose

$$M_{R4} = M_{z4}. \tag{44}$$

The remaining even fourth moments then follow from Eqs. (41)–(43). All six remain nonnegative over the active tracer domain and satisfy the corresponding Cauchy moment inequalities.

For LOS direction cosines $(\ell_R, \ell_\phi, \ell_z)$,

$$\overline{v_{\mathrm{los}}^4} = \ell_R^4 M_{R4} + \ell_\phi^4 M_{\phi 4} + \ell_z^4 M_{z4} \tag{45}$$

$$+ 6\ell_R^2\ell_\phi^2 M_{R2\phi 2} + 6\ell_R^2\ell_z^2 M_{R2z2} + 6\ell_\phi^2\ell_z^2 M_{\phi 2z2}. \tag{46}$$

At the source-assigned 1.3 kpc comparison coordinate this minimal continuation gives

$$\kappa_{\mathrm{CPTG}} = -0.15167. \tag{47}$$

A near-Gaussian moment mapping would correspond to $h_4^{\mathrm{eq}} \simeq -0.00774$, but that mapping is used only as a small-deviation diagnostic and not as an exact conversion of the measured $h_4 = 0.13$. The robust statement is the sign: the minimal closure produces slight negative excess kurtosis and therefore does not account for the observed positive line-profile shape.

For an even two-integral distribution $f(E, L_z)$, rotational symmetry in the (v_R, v_z) velocity plane enforces

$$\overline{v_R^2} = \overline{v_z^2}, \qquad \overline{v_R^4} = \overline{v_z^4}, \qquad \overline{v_R^2 v_z^2} = \frac{1}{3}\overline{v_R^4}. \tag{48}$$

These identities show that a two-integral continuation does not supply independent meridional fourth-order freedom [8]. A three-integral orbital population therefore supplies the required additional dynamical freedom.

10.2 Three-integral orbit-superposition consistency analysis

A Schwarzschild-style orbit library is constructed directly in the fixed conservative CPTG potential [9]. Each numerical trajectory is represented together with equal-weight time-reversed and mid-plane-reflected symmetry partners, preserving zero net streaming and cancellation of the meridional cross term. The retained orbit weights satisfy $w_j \geq 0$.

The intrinsic source is divided into 32 meridional cells. In every cell the orbit superposition is required to reproduce the tracer mass and the numerators associated with σ_R^2, σ_z^2, and $\overline{v_\phi^2}$. The same weights also reproduce the nine projected-light constraints, nine projected second-moment numerators, the light and second-moment numerator in the 1.3 kpc comparison interval, and total mass normalization. This gives 149 upstream scaled constraints before any h_4 information is supplied.

For orbit library A, 1020 of 1024 trajectories pass the domain and conservation gates. The Hamiltonian drift is

$$\text{median}(\max|\Delta E/E|) = 4.85 \times 10^{-5}, \qquad P_{95} = 2.26 \times 10^{-4}, \tag{49}$$

with maximum drift 8.02×10^{-4} and maximum relative angular-momentum drift

$$\max|\Delta L_z/L_z| = 9.90 \times 10^{-13}. \tag{50}$$

An independent 1024-orbit seed also retains 1020 trajectories, with $P_{95}(\max|\Delta E/E|) = 2.31 \times 10^{-4}$ and maximum relative L_z drift 6.18×10^{-13}.

The LOSVD in the comparison interval is accumulated directly from the weighted orbit velocities and fitted with the fourth-order Gauss–Hermite basis used for the observational comparison [6]. With h_4 withheld, the regularized nonnegative solutions give

$$h_{4,\text{withheld}}^{(1)} = -0.00416, \qquad h_{4,\text{withheld}}^{(2)} = -0.02164. \tag{51}$$

Thus the positive observed coefficient does not emerge automatically from a neutral orbit weighting.

The stronger question is whether the observed LOSVD shape is dynamically admissible in the unchanged CPTG potential. Supply the measured $h_4 = 0.13$ only at the final orbit-weighting stage by constraining the radius-matched Gauss–Hermite LOSVD shape while leaving the gravitational field and all 149 upstream constraints unchanged. Orbit library A then gives

$$h_{4,\text{direct}}^{(1)} = 0.13000299, \qquad \max|\Delta_{\text{upstream}}| = 2.35 \times 10^{-7}, \tag{52}$$

and the independent library gives

$$h_{4,\text{direct}}^{(2)} = 0.13000175, \qquad \max|\Delta_{\text{upstream}}| = 6.87 \times 10^{-7}. \tag{53}$$

The corresponding effective orbit numbers are approximately 595 and 612, so the solutions are not supported by only a few exceptional trajectories.

The orbit-superposition result is numerically stable. Halving the leapfrog time step gives $h_4 = 0.12999934$; increasing the intrinsic meridional grid from 32 to 40 cells gives $h_4 = 0.13000016$; and changing the comparison half-width from 0.10 kpc to 0.05 and 0.20 kpc gives $h_4 = 0.12998633$ and 0.13000369, respectively. Every case retains nonnegative weights and upstream scaled residuals below 7×10^{-6}.

Orbit library	Retained orbits	max upstream residual	Direct fitted h_4
Library A	1020/1024	2.35×10^{-7}	0.130003
Independent seed	1020/1024	6.87×10^{-7}	0.130002
Half time step	1020/1024	1.55×10^{-6}	0.129999
40 intrinsic cells	1020/1024	1.18×10^{-6}	0.130000
0.05 kpc half-width	1020/1024	6.84×10^{-6}	0.129986
0.20 kpc half-width	1020/1024	4.78×10^{-7}	0.130004

Table 6: Three-integral LOSVD compatibility and numerical sensitivity in the same fixed CPTG potential. The observed h_4 is supplied only in the final orbit-weighting stage, so these rows establish dynamical compatibility rather than an independent prediction.

A separate maximum-entropy selection evaluates whether a neutral prior predicts the observed LOSVD shape without using h_4 [10]. Maximizing entropy subject only to the 149 upstream constraints gives

$$h_{4,\mathrm{ME}}^{(1)} = -0.00063, \qquad h_{4,\mathrm{ME}}^{(2)} = -0.01823. \tag{54}$$

The corresponding maximum scaled upstream residuals are 1.74×10^{-6} and 2.99×10^{-7}. The observed positive h_4 is therefore allowed by the fixed CPTG equilibrium orbit space but is not selected by this neutral entropy prescription. Predictive higher-order closure remains open until CPTG supplies an independently derived rule for the third-integral orbit population.

11 Structural Mode of the Solved Field

Structural Mode N is a compact, dimensionless description of the radial organization of a solved CPTG gravitational field [12]. Its purpose is not to measure the total strength of gravity or the quality of the kinematic fit, and it is not a visual morphology label. Instead, it asks where the field's structural variation is concentrated within the resolved outer domain. The construction forms a field invariant from $g(r)$ and its radial gradient, weights each radius by how rapidly that invariant changes, and computes the weighted radial centroid λ of that change. The ratio $N = R/\lambda$ then expresses that organization relative to the outer solved radius R: smaller N places the dominant structural variation farther outward, while larger N places it progressively farther inward. In this way, N can expose organized gravitational structure that is not evident from surface brightness alone.

Structural Mode is evaluated only after the Dragonfly 44 gravitational field and stellar dynamical solution are fixed. It is therefore strictly downstream of the dynamical inference and does not enter the determination of $a_\star$, the conservative potential, the Jeans stresses, the orbital library, or the orbit weights. For the continuous solved field, define

$$C_g(r) = g(r)^2 + \kappa_{\rm struct}^2 \left(\frac{dg}{dr}\right)^2, \qquad \kappa_{\rm struct} = 1\,{\rm kpc}, \tag{55}$$

with structural weight

$$w(r) = \left|\frac{dC_g}{dr}\right|. \tag{56}$$

On the resolved outer coordinate, the parent continuum readout is

$$\lambda = \frac{\int_{R/5}^{R} r\,w(r)\,dr}{\int_{R/5}^{R} w(r)\,dr}, \qquad N = \frac{R}{\lambda}. \tag{57}$$

Using

$$R = \frac{5.13}{0.83} = 6.1807228916\,{\rm kpc}, \qquad \lambda = 3.7324869990\,{\rm kpc}, \tag{58}$$

gives

$$N_{\rm DF44} = 1.6559261675. \tag{59}$$

12 Interpretation and Claim Boundary

The Dragonfly 44 calculation is not a parameter-free prediction from photometry alone. Equation (30) uses the observed integrated half-light support to determine one object-level structural acceleration. The resolved nine-bin comparison asks whether that one structural state, propagated through a conservative axisymmetric field and the stellar stress tensor, reproduces the observed radial structure without independent radial adjustment. The primary profile is conditional on the central $33\,{\rm km\,s^{-1}}$ half-light value; Table 2 shows that repeating the full normalization at the quoted 30 and $36\,{\rm km\,s^{-1}}$ endpoints leaves the largest pointwise discrepancy below 2.46σ. Because the integrated half-light measurement overlaps the resolved spectroscopy, these endpoint calculations are treated as sensitivity bounds rather than independent likelihood contributions.

The Structural Mode result is downstream of that dynamical calculation. The value $N_{\rm DF44} = 1.655926168$ describes the radial organization of the already solved field and does not repair or select any upstream model quantity.

The higher-order analysis sharpens that boundary. Section 10 shows that the minimal fourth-order/two-integral continuation does not generate the observed positive LOSVD shape. A three-integral orbit-superposition calculation in the same fixed CPTG potential does admit the measured h_4 with nonnegative weights while preserving the discretized three-dimensional tracer density, intrinsic second moments, projected light, and projected second moments. This is a dynamical compatibility result, not an untuned prediction, because h_4 is supplied only at the final orbit-weighting stage. Maximum-entropy weights selected without h_4 remain near zero or mildly negative.

Finally, Eq. (18) is a parameter-free axisymmetric CPTG reduction. It is mathematically conservative and retains the equatorial CPTG scalar response, but it is not presented as the arbitrary-geometry component expression of the complete parent $\Theta^{\mathrm{trans}}_{\mu\nu}$ sector.

13 Conclusion

With the source-stated radial conversion, circularized half-light normalization, conservative CPTG field, and full projected tensor observation operator, Dragonfly 44 is consistent with the CPTG galaxy response used here. The high-resolution edge-on neutral solution gives

$$a_\star = 5.2935 \times 10^{-10}\ \mathrm{m\,s^{-2}}, \qquad R_c = 829.73\ \mathrm{Gpc}, \tag{60}$$

and, for the central $33\,\mathrm{km\,s^{-1}}$ half-light normalization, all nine resolved KCWI measurements lie within 1.47σ. The descriptive radial statistic is

$$\chi^2_{\mathrm{diag}} = 9.21. \tag{61}$$

As a downstream structural readout of the same fixed continuous field,

$$N_{\mathrm{DF44}} = 1.655926168. \tag{62}$$

Dragonfly 44 is conventionally interpreted as a “failed galaxy,” but the Structural Mode result gives a different description of its present internal state. Within CPTG, the solved baryon-sourced field is coherent and well organized despite the galaxy’s extreme diffuseness. Dragonfly 44 is therefore low in surface brightness without being structurally disordered: the “failed” designation describes the conventional halo-based formation history, not the internal gravitational organization recovered here [1, 2].

The identical stellar-only tensor calculation gives

$$\chi^2_{\star,\mathrm{diag}} = 539.66, \tag{63}$$

showing that the successful comparison is not produced by axisymmetric tensor geometry alone. Repeating the whole-galaxy normalization at the 30 and $36\,\mathrm{km\,s^{-1}}$ endpoints gives maximum resolved pulls of 1.16σ and 2.45σ, respectively.

The second-moment result is stable across inclination, meridional-stress, stellar-mass, spatial-resolution, source-reprojection, field-integrability, and numerical-convergence variations without a dark-halo source, stellar-source weight, pointwise $a_\star$, fitted Structural Mode, Satoh amplitude, or continuously optimized anisotropy parameter. For non-circular stellar systems, the relevant CPTG second-moment comparison is therefore the projected stress tensor generated by the stated geometry, rather than a scalar σ^2/r surrogate.

The higher-order continuation separates restrictive closure failure from gravitational-field viability. The minimal fourth-order/two-integral diagnostic produces slight negative excess kurtosis and does not account for the observed positive h_4. In contrast, two independently seeded three-integral orbit libraries in the unchanged CPTG potential each admit broad nonnegative solutions whose directly fitted radius-matched Gauss–Hermite LOSVDs reproduce $h_4 = 0.13$ while keeping all 149 upstream density and second-moment constraints at sub-10^{-6} scaled residuals. The result is stable to time-step, intrinsic-cell, and comparison-window changes. The measured LOSVD shape is therefore dynamically compatible with the solved CPTG potential. It is not yet an untuned prediction: maximum-entropy orbit weights chosen without the h_4 datum remain near zero or mildly negative, so an independently derived third-integral population law is still required for predictive LOSVD closure.

Data, Code, and Evidence Availability

The Dragonfly 44 photometric and kinematic inputs are drawn from van Dokkum et al. [3]. The accompanying numerical evidence archive is

`CPTG_Dragonfly44_COMPLETE_PUBLICATION_EVIDENCE_20260910.zip`

It contains the deterministic second- and fourth-moment solvers, observation-operator record, nine-bin comparison, stellar-only control, half-light normalization sensitivity, inclination/stress and stellar-mass grids, radial-coordinate sensitivity, numerical-convergence and integrability calculations, central-force limiting diagnostic, three-integral orbit solver, orbit libraries, intrinsic and projected constraint residuals, Hamiltonian and angular-momentum conservation records, time-step/cell/window sensitivity calculations, maximum-entropy ledgers, the exact Structural Mode parent-domain calculation, replay commands, runtime records, and a complete SHA-256 manifest. Conventional halo parameters are not used as CPTG source inputs.

Supporting CPTG theory resources are maintained at: `https://github.com/CLG2025/CPTG`

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